University of Hertfordshire UH

Logbook 4

Name:	**Guruprasad**
Week Date:	**Part A -Meeting 4 of 8**

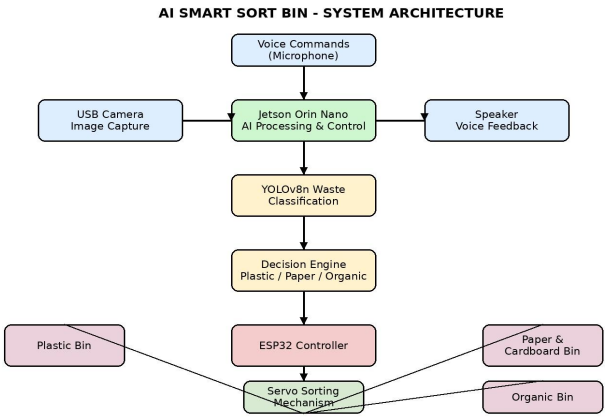
(*The above information is for you to keep track of which week this log is applicable for. This information is not required in the online logbook as the system will find your name, your supervisor's name, and take the submission date, automatically. You must submit the online log in the relevant week. Logs* ***cannot*** *be backdated to missed weeks.)*

Meeting date: *(please note if no meeting is planned this week)*	**Date and day** **23/7/2026, Thursday**
Work Undertaken this week: *Not more than 500 words*	**From the last meeting until now, what have you done based on the task you have listed in the last logbook?** **My Project is AI-Enabled Smart Waste classification and Sorting System (Smart Bin)** **I have completed draft project outline and have modified after discussion with Mr.Wong, Project Head coordinator, I also prepared resource check list. I have Completed Risk Assessment form. Prepared System Architecture. Prepared bill of material for Hardware and software. Major components procured. All will be attached as annexure to project outline document.**
Problems encountered: *(Not more than 500 words*	**What are the tasks that are still outstanding to write your Project Outline** **Planning to add wiring diagrams, developing various layers like data acquisition, Voice Interaction are bit challenging, working on the same. Also deployment of Jetson Nano**
Tasks agreed for upcoming week before the next meeting	**What are your plan actions/task until the next progress meeting that are agreed with your supervisor** **Now I plan to start data gathering, dust bin fabrication drawing and detailed wiring drawings and coding for various layers like AI processing, Voice recognition and interaction layer etc** **Planning edge deployment also as I have sourced Jetson Orin Nano.**

General

Figures:
(Upload image files in the online Log Form, but feel free to keep notes here about which images you intend to upload. Use "Figure Caption" information as the filenames, e.g., "Figure 1 – A screenshot of the initial constrained Sketch in CATIA.jpg")

1) Update Hardware Block diagram of your system



2) Key software codes/modules that need to be developed.

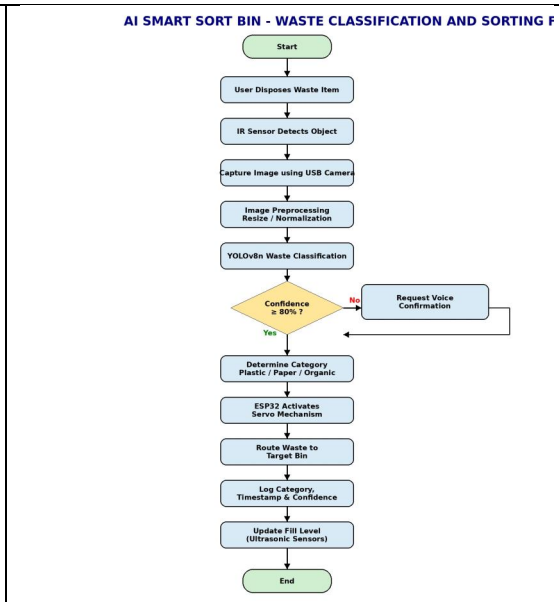
- **Data processing**
- **Voice interaction**
- **AI processing**
- **Classification logic**
- **Edge deployment**
- **Data storage**

All details are part of project outline document that are in Section 2

3) Any Quality and Regulatory issues addressed?

Mechanical and fabrication issues
Dataset availability(quality)
Environment condition
Voice recognition constraints
Risk analysis done for all constraints and action plan is added in the project outline as Annexure E

4) A updated/improved version of your system sketch


AI SMART SORT BIN - WASTE CLASSIFICATION AND SORTING F
Start
User Disposes Waste Item
IR Sensor Detects Object
Capture Image using USB Camera
Image Preprocessing
Resize / Normalization
YOLOv8n Waste Classification
Confidence
≥ 80% ?
No
Request Voice
Confirmation
Yes
Determine Category
Plastic / Paper / Organic
ESP32 Activates
Servo Mechanism
Route Waste to
Target Bin
Log Category,
Timestamp & Confidence
Update Fill Level
(Ultrasonic Sensors)
End